

Econometric Analysis of Variance Inflation and Convex Hull Violations under the Alternative Hypothesis

Research Note

June 19, 2026

This note provides a rigorous econometric analysis of the mathematical and algorithmic reasons for the power collapse and numerical failures observed in the two-stage Orthogonalized Sieve Empirical Likelihood (Sieve-EL) framework under the alternative hypothesis $H_1 : \beta \neq \beta_0$.

1 The Convex Hull Violation Pathology

1.1 Mathematical Mechanism

In standard Empirical Likelihood (EL), the moment condition evaluated at the null hypothesis value β_0 is defined as:

$$Z_t(\beta_0) = \left(\tilde{Y}_t - \beta_0 X_{t-1} \right) w_t^c \quad (1)$$

where $\tilde{Y}_t = Y_t - \mathbf{P}_K(W_{t-1})'\hat{\mathbf{c}}$ is the response adjusted by the first-stage nonparametric sieve estimator, and w_t^c is the orthogonalized weight.

Let the true data-generating process (DGP) under the alternative hypothesis be:

$$Y_t = m(W_{t-1}) + \beta X_{t-1} + U_t, \quad \text{with } \beta \neq \beta_0 \quad (2)$$

Using the sieve decomposition $m(W_{t-1}) = \mathbf{P}_K(W_{t-1})'\mathbf{c} + r_t$, the adjusted return is:

$$\tilde{Y}_t = \mathbf{P}_K(W_{t-1})'\mathbf{c} + r_t + \beta X_{t-1} + U_t - \mathbf{P}_K(W_{t-1})'\hat{\mathbf{c}} \quad (3)$$

Substituting this into the moment condition yields:

$$Z_t(\beta_0) = (U_t + r_t - \mathbf{P}_K(W_{t-1})'(\hat{\mathbf{c}} - \mathbf{c}) + (\beta - \beta_0)X_{t-1}) w_t^c \quad (4)$$

Since the weight orthogonalization in Step 4 ensures that $\sum_{t=1}^T \mathbf{P}_K(W_{t-1})w_t^c \equiv \mathbf{0}$ in-sample, the sieve estimation error $(\hat{\mathbf{c}} - \mathbf{c})$ is annihilated when summing the moments. However, for individual observations, the moment is dominated by the signal:

$$Z_t(\beta_0) \approx U_t w_t^c + (\beta - \beta_0)X_{t-1} w_t^c \quad (5)$$

1.2 The Convex Hull Condition

The weight function is defined as $w(X_{t-1}) = \tanh(X_{t-1}/c_w)$. Because $\tanh(x)$ is an odd function that preserves the sign of x , the product $X_{t-1}w(X_{t-1})$ is strictly positive for all $X_{t-1} \neq 0$:

$$X_{t-1}w(X_{t-1}) = X_{t-1} \tanh\left(\frac{X_{t-1}}{c_w}\right) \geq 0 \quad (6)$$

Since X_{t-1} and W_{t-1} are independent, the projection coefficient $\hat{\gamma} \approx \mathbf{0}$ asymptotically, meaning $w_t^c \approx w(X_{t-1})$. Consequently:

$$X_{t-1}w_t^c \approx X_{t-1} \tanh\left(\frac{X_{t-1}}{c_w}\right) > 0 \quad \text{a.s.} \quad (7)$$

Under the alternative hypothesis, the moment is shifted by the signal term:

$$\mathbb{E}[Z_t(\beta_0)] \approx (\beta - \beta_0)\mathbb{E}[X_{t-1}w_t^c] \quad (8)$$

If $\beta > \beta_0$, then $\mathbb{E}[Z_t(\beta_0)] > 0$. If $\beta < \beta_0$, then $\mathbb{E}[Z_t(\beta_0)] < 0$.

In finite samples, as the deviation $|\beta - \beta_0|$ or the sample size T grows, the deterministic signal term $(\beta - \beta_0)X_{t-1}w_t^c$ dominates the zero-mean innovation term $U_t w_t^c$. This causes all sample moments to have the same sign:

$$\text{sign}(Z_t(\beta_0)) = \text{sign}(\beta - \beta_0) \quad \forall t \in \{1, \dots, T\} \quad (9)$$

When this occurs, the convex hull of the set $\{Z_t(\beta_0)\}_{t=1}^T$ does not contain zero:

$$0 \notin \text{Conv}(\{Z_t(\beta_0)\}_{t=1}^T) \quad (10)$$

1.3 Algorithmic Consequence

Under standard EL theory, $0 \notin \text{Conv}(\{Z_t\})$ implies that the null hypothesis is rejected with certainty, corresponding to $\ell_T(\beta_0) = \infty$.

However, in numerical implementations, this violation causes the root-finding algorithm (e.g., `brentq` or `scipy.optimize`) to fail since the gradient does not bracket zero. The code returns a penalty value of 10^6 , which is then filtered out as a “numerical failure” in `diagnostic_chisq_test.py`. Filtering out these rejections artificially drops the most powerful test statistics from the Monte Carlo simulation, leading to a flat and severely distorted empirical power curve.

2 The Variance Inflation Pathology

Even in cases where 0 is in the convex hull (e.g., for very small deviations or small sample sizes), the test statistic suffers from severe deflation.

2.1 Asymptotic Analysis of the EL Statistic

By Taylor-expanding the profile EL statistic $\ell_T(\beta_0)$, we obtain:

$$\ell_T(\beta_0) \approx \frac{\left(\frac{1}{\sqrt{T}} \sum_{t=1}^T Z_t(\beta_0)\right)^2}{\frac{1}{T} \sum_{t=1}^T Z_t(\beta_0)^2} \quad (11)$$

Evaluating the terms under the alternative hypothesis:

1. Numerator (Score-like term):

$$\frac{1}{\sqrt{T}} \sum_{t=1}^T Z_t(\beta_0) = \frac{1}{\sqrt{T}} \sum_{t=1}^T U_t w_t^c + (\beta - \beta_0) \frac{1}{\sqrt{T}} \sum_{t=1}^T X_{t-1} w_t^c + o_p(1) \quad (12)$$

The signal term grows at rate $O_p(\sqrt{T}(\beta - \beta_0) \cdot \text{std}(X))$.

2. Denominator (Sample Variance of moments):

$$\hat{V}_T = \frac{1}{T} \sum_{t=1}^T Z_t(\beta_0)^2 \approx \frac{1}{T} \sum_{t=1}^T U_t^2 (w_t^c)^2 + (\beta - \beta_0)^2 \frac{1}{T} \sum_{t=1}^T X_{t-1}^2 (w_t^c)^2 \quad (13)$$

Because the moment condition $Z_t(\beta_0)$ is evaluated at the null value β_0 , the signal term $(\beta - \beta_0)X_{t-1}$ is not removed from the residuals. Thus, the sample variance is inflated by the term $(\beta - \beta_0)^2 \frac{1}{T} \sum_{t=1}^T X_{t-1}^2 (w_t^c)^2$.

Because X_{t-1} is highly persistent ($\rho \approx 1$), its variance is extremely large, which severely inflates the denominator.

2.2 Power Collapse

As the true parameter β moves further from the null β_0 , the signal terms dominate both the numerator and the denominator. Taking the limit as $|\beta - \beta_0| \rightarrow \infty$:

$$\lim_{|\beta - \beta_0| \rightarrow \infty} \ell_T(\beta_0) \approx \frac{\left(\frac{1}{\sqrt{T}} \sum_{t=1}^T X_{t-1} w_t^c\right)^2}{\frac{1}{T} \sum_{t=1}^T X_{t-1}^2 (w_t^c)^2} = \frac{\left(\sum_{t=1}^T X_{t-1} w_t^c\right)^2}{\sum_{t=1}^T X_{t-1}^2 (w_t^c)^2} \quad (14)$$

By the Cauchy-Schwarz inequality, this ratio is bounded:

$$\ell_T(\beta_0) \leq \sum_{t=1}^T (1)^2 = T \quad (15)$$

In practice, since $w_t^c \approx \tanh(X_{t-1}/c_w)$, the ratio is bounded by a much smaller constant than T . As a result, the EL statistic plateaus and cannot diverge to infinity. The test is unable to consistently detect large deviations from the null, leading to a catastrophic loss of statistical power.

3 The Proposed Solution

To restore the power of the test, we must decouple the variance estimator in the denominator from the signal term $(\beta - \beta_0)X_{t-1}$. We replace the standard EL profile optimization with a score-type test statistic that uses the **unconstrained residuals** to estimate the variance.

1. Estimate the unconstrained model parameters $(\hat{\mathbf{c}}, \hat{\beta}, \hat{\phi})$ via joint OLS:

$$Y_t = \mathbf{P}_K(W_{t-1})'\mathbf{c} + \beta X_{t-1} + \phi\hat{\varepsilon}_t + \text{error}_t \quad (16)$$

2. Compute the unconstrained residuals:

$$\hat{U}_t = Y_t - \mathbf{P}_K(W_{t-1})'\hat{\mathbf{c}} - \hat{\beta}X_{t-1} - \hat{\phi}\hat{\varepsilon}_t \quad (17)$$

Since $\hat{\beta} \xrightarrow{p} \beta$ under both the null and the alternative, the unconstrained residuals consistently estimate the pure innovations z_t : $\hat{U}_t \xrightarrow{p} z_t$.

3. Construct the unconstrained variance estimator:

$$\hat{V}_{unconstrained} = \frac{1}{T} \sum_{t=1}^T \hat{U}_t^2 (w_t^c)^2 \quad (18)$$

This estimator is strictly locked to the innovation variance and does not suffer from variance inflation under the alternative.

4. Define the robust score-type test statistic:

$$S_T(\beta_0) = \frac{\left(\frac{1}{\sqrt{T}} \sum_{t=1}^T Z_t(\beta_0) \right)^2}{\hat{V}_{unconstrained}} \quad (19)$$

Under H_0 , $S_T(\beta_0) \xrightarrow{d} \chi_1^2$. Under H_1 , the denominator remains $O_p(1)$ while the numerator grows at rate $O_p(T(\beta - \beta_0)^2)$, restoring 100% asymptotic power.